



Spring MVC with Boot

- ♦ You can develop Spring MVC application with Spring Boot in two ways.
 - Develop as **FAT JAR** file.
 - Develop as **WAR** file
- ♦ FAT Jar is self executable application which contains **Application Code + Embbeded Container**
- ♦ You can execute the FAT jar as follows.
 - Java -jar HelloApp.jar
- ♦ WAR file is deployable application which can be deployed in any External Container to run.

demo1-mvc-boot: Spring MVC with Boot (using JSP)

Working Steps:

A) Setup the Project

1) Create the maven Project.

- a) Open the Eclipse required Workspace
- b) Select New -> Maven Project.
- c) Check the checkbox for create Simple Project(skip the Archetype selection) and Click on Next Button.
- d) Provide the following
 - Group Id: com.coursecube.springboot
 - Artifact Id : **demo1-mvc-boot**
 - Version : 1.0
 - packaging : jar
 - Name : **demo1-mvc-boot**and Click On finish button.

2) Remove the following two Source Folders.

- a. src/test/java
- b. src/test/resources

3) Create application.properties under src/main/resources

4) Update application.properties as follows

server.port=12345

5) Create the Source Folder called **src/main/webapp**

6) Create the Folder called **WEB-INF** under src/main/webapp



7) Create the Folder called **myjsps** under WEB-INF

8) Update pom.xml by adding the following dependencies.

```
<parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>2.1.16.RELEASE</version>
</parent>

<dependencies>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
        <groupId>org.apache.tomcat.embed</groupId>
        <artifactId>tomcat-embed-jasper</artifactId>
    </dependency>

    <dependency>
        <groupId>jstl</groupId>
        <artifactId>jstl</artifactId>
        <version>1.2</version>
    </dependency>

    <dependency>
        <groupId>>taglibs</groupId>
        <artifactId>standard</artifactId>
        <version>1.1.2</version>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-devtools</artifactId>
    </dependency>
</dependencies>
```

9) You can see the Error in Maven Project.

Fix the Error by updating the Maven Project.

Select the Project , right click and then select Maven -> Update Project.



- 10) Create the Package called **com.jlcindia.springboot** under src/main/java
- 11) Write Bean Configuration class – **JLCWebConfig.java**
- 12) Configure View Resolver in **JLCWebConfig.java**

@SpringBootApplication

```
public class JLCWebConfig implements WebMvcConfigurer{
```

@Bean

```
    public ViewResolver viewResolver() {  
        InternalResourceViewResolver resolver = new InternalResourceViewResolver();  
        resolver.setPrefix("/WEB-INF/myjsps/");  
        resolver.setSuffix(".jsp");  
        resolver.setViewClass(JstlView.class);  
        return resolver;  
    }  
}
```

- 13) Write Boot Main Class – MyBootApplication.java

@SpringBootApplication

```
public class MyBootApplication {  
    public static void main(String[] args) {  
        SpringApplication.run(StartMyBootApplication.class, args);  
    }  
}
```

- 14) Run the Boot Application

You will see – Whitelabel Error Page means DS is Ready.

B)Develop the Usecase : show index page

http://localhost:12345/ => You are sending the request for /

- 15) Write index.jsp under WEB-INF/myjsps folder.

```
<!DOCTYPE html>  
<html>  
<body>  
<h1> Welcome to Java Learning Center !!!</h1>  
</body>  
</html>
```



16) Write IndexController.java

```
@Controller
public class IndexController {
    @GetMapping("/")
    public String showIndexPage() {
        System.out.println("IndexController - showIndexPage()");
        return "index";
    }
}
```

17) Run the Boot Application

18) Open the browser and hit <http://localhost:12345/>
If every thing fine you will see index.jsp

C)Enable BootStarp:

19) Specify the Dependencies for JQuery and Bootstarp

```
<dependency>
    <groupId>org.webjars</groupId>
    <artifactId>bootstrap</artifactId>
    <version>4.3.1</version>
</dependency>

<dependency>
    <groupId>org.webjars</groupId>
    <artifactId>jquery</artifactId>
    <version>3.4.1</version>
</dependency>
```

20) Observe the following

a) bootstrap-4.3.1.jar

META-INF=> resources => webjars => bootstrap => 4.3.1 => css => bootstrap.min.css

META-INF=> resources => webjars => bootstrap => 4.3.1 => js => bootstrap.min.js

b) jquery-3.4.1.jar =>

META-INF=> resources => webjars => jquery => 3.4.1 => jquery.min.js



- 21) Add the resource handler bean (to make your resources available in class path) in JLCWebConfig

```
@Override
public void addResourceHandlers(ResourceHandlerRegistry registry) {
    registry.addResourceHandler("/webjars/**").addResourceLocations("classpath:/META-
-INF/resources/webjars/");
}
```

- 22) Include the css and js files in index.jsp.

```
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>
```

- 23) Use Some Bootstrap Classes in index.jsp

```
<h1 class="text-center"> Welcome to Java Learning Center !!!</h1>
```

- 24) Run the Boot Application and Test

- 25) Open the browser and hit <http://localhost:12345/>

If every thing fine you will see Bootstrap Css styles applied in index.jsp

D)Develop the Usecase : show books

<http://localhost:12345/showBooks>

- 26) Add hyperlink in index.jsp

```
<h2 class="text-center"> <a href="showBooks"> Show Books </a></h2>
```

- 27) Write Book.java under **com.jlcindia.springboot**

- 28) Write BooksController.java under **com.jlcindia.springboot**

- 29) Write booksList.jsp under **WEB-INF/myjsps**

- 30) Run the Boot Application and Test

Open the browser and hit <http://localhost:12345/>

demo1-mvc-boot: Files Required

1. index.jsp	2. booksList.jsp
3. IndexController.java	4. BooksController.java
5. Book.java	6. JLCWebConfig.java
7. MyBootApplication.java	8. application.properties
9. pom.xml	

Welcome to CourseCube !!!

Latest Books from Srinivas Dande

Book Id	Book Name	Author	Price	Publications		
101	Master Spring Boot	Srinivas Dande	10000.0	JLC	View Book	Edit Book
102	Master MicroServices	Srinivas Dande	10000.0	JLC	View Book	Edit Book
103	Master Angular	Srinivas Dande	10000.0	JLC	View Book	Edit Book
104	Master React	Srinivas Dande	10000.0	JLC	View Book	Edit Book
105	Master Java Full Stack	Srinivas Dande	10000.0	JLC	View Book	Edit Book

1) index.java

```

<!DOCTYPE html>
<html>
<head>
<title> JLCBookStore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<h1 class="text-center"> Welcome to Java Learning Center !!!</h1>
<br/>
<h2 class="text-center"> <a href="showBooks"> Show Books </a></h2>

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

</body>
</html>

```



2) BooksList.jsp

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>

<!DOCTYPE html>
<html>  <head>
<title> JLCBookStore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="container">
<h1 class="text-center"> Welcome to Java Learning Center !!!</h1>

<div class="alert alert-danger" role="alert">
<h2 class="text-center"> Latest Books from Srinivas Dande </h2>
</div>

<br/>
<table class="table table-striped table-bordered table-hover" style="font-size:20px;">
<thead class="thead-light">
<tr>  <th> Book Id </th>
<th> Book Name </th>
<th> Author</th>
<th> Price </th>
<th> Publications</th>
<th> </th>
<th> </th>  </tr>
</thead>
<tbody>
<c:forEach var="mybook" items="{MyBooks}">
<tr>
<td> <a href="#"> ${mybook.bid} </a> </td>
<td> ${mybook.bname} </td>
<td> ${mybook.author} </td>
<td> ${mybook.price} </td>
<td> ${mybook.pub}</td>
<td> <button type="button" class="btn btn-success btn-lg">View Book</button> </td>
<td> <button type="button" class="btn btn-primary btn-lg">Edit Book</button> </td>
</tr>
</c:forEach>
</tbody>
</table>
</div>
<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>
</body>
</html>
```



3) IndexController.java

```
package com.jlcindia.springboot;

import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Controller
public class IndexController {

    @GetMapping(value = "/")
    public String showIndexPage() {
        System.out.println("IndexController - showIndexPage()");
        return "index";
    }
}
```

4) BooksController.java

```
package com.jlcindia.springboot;

import java.util.ArrayList;
import java.util.List;
import javax.servlet.http.HttpSession;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Controller
public class BookController {

    @GetMapping("/showBooks")
    public String getBooks(HttpSession session) {
        System.out.println(" BooksController - getBooks()");

        List<Book> blist=new ArrayList<>();

        Book mybook1=new Book(101,"Master Spring Boot","Srinivas Dande ",10000.0,"JLC");
        Book mybook2=new Book(102,"Master MicroServices","Srinivas Dande",10000.0,"JLC");
        Book mybook3=new Book(103,"Master Angular","Srinivas Dande",10000.0,"JLC");
        Book mybook4=new Book(104,"Master React ","Srinivas Dande",10000.0,"JLC");
        Book mybook5=new Book(105,"Master Java Full Stack","Srinivas Dande",10000.0,"JLC");
    }
}
```



```
blist.add(mybook1);
blist.add(mybook2);
blist.add(mybook3);
blist.add(mybook4);
blist.add(mybook5);

session.setAttribute("MyBooks", blist);

return "booksList";
}
}
```

5) Book.java

```
package com.jlcindia.springboot;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

public class Book {

    private int bid;
    private String bname;
    private String author;
    private double price;
    private String pub;

    public Book() {}
    public Book(int bid, String bname, String author, double price, String pub) {
        super();
        this.bid = bid;
        this.bname = bname;
        this.author = author;
        this.price = price;
        this.pub = pub;
    }
    //Setters and Getters
    //toString()
}
```



6) JLCWebConfig.java

```
package com.jlcindia.springboot;

import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.Bean;
import org.springframework.web.servlet.ViewResolver;
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
import org.springframework.web.servlet.view.InternalResourceViewResolver;
import org.springframework.web.servlet.view.JstlView;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class JLCWebConfig implements WebMvcConfigurer{
    @Bean
    public ViewResolver viewResolver() {
        InternalResourceViewResolver resolver = new InternalResourceViewResolver();
        resolver.setPrefix("/WEB-INF/myjsps/");
        resolver.setSuffix(".jsp");
        resolver.setViewClass(JstlView.class);
        return resolver;
    }
    @Override
    public void addResourceHandlers(ResourceHandlerRegistry registry) {
        registry.addResourceHandler("/webjars/**")
        .addResourceLocations("classpath:/META-INF/resources/webjars/");
    }
}
```

7) MyBootApplication.java

```
package com.jlcindia.springboot;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication {
    public static void main(String as[]) {
        SpringApplication.run(MyBootApplication.class, as);
    }
}
```



8)application.properties

```
server.port=12345
```

9)pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo1-mvc-boot</artifactId>
<version>1.0</version>
<name>demo1-mvc-boot</name>
<description>This is My Spring MVC App with Boot</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.apache.tomcat.embed</groupId>
<artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>
```



```
<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>myjlcapp</finalName>
<plugins>

<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```



demo2-mvc-boot: Spring MVC with Boot (using ThymeLeaf)

Working Steps:

- 1) Copy **demo1-mvc-boot** as **demo2-mvc-boot**
- 2) Remove the WEB-INF under src/main/webapp (So that all the JSP's will be deleted)
- 3) Remove **ViewResolver** Bean from JLCWebConfig.java
- 4) Create templates folder under src/main/resources
- 5) Place index.html and booksList.html under templates
 - a) index.html (new in Demo2)
 - b) booksList.html (new in Demo2)
- 6) Add the following property in application.properties

```
server.port=54321
spring.thymeleaf.cache=false
```
- 7) Remove the following dependencies from pom.xml

```
<dependency>
  <groupId>org.apache.tomcat.embed</groupId>
  <artifactId>tomcat-embed-jasper</artifactId>
</dependency>

<dependency>
  <groupId>jstl</groupId>
  <artifactId>jstl</artifactId>
  <version>1.2</version>
</dependency>

<dependency>
  <groupId>>taglibs</groupId>
  <artifactId>standard</artifactId>
  <version>1.1.2</version>
</dependency>
```
- 8) Add the dependency for ThymeLeaf in pom.xml

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-thymeleaf</artifactId>
</dependency>
```

- 9) Make sure that all the Files as given below are updated correctly.
- 10) Run the Boot Application
- 11) Open the browser and hit <http://localhost:54321/>

demo2-mvc-boot: Files Required

1. index.html	New in Demo2
2. booksList.html	New in Demo2
3. Book.java	Same as Demo1
4. IndexController.java	Same as Demo1
5. BooksController.java	Same as Demo1
6. JLCWebConfig.java	Updated in Demo2
7. StartMyBootApplication.java	Same as Demo1
8. application.properties	Updated in Demo2
9. pom.xml	Updated in Demo2

Welcome to CourseCube !!!

Latest Books from Srinivas Dande

Book Id	Book Name	Author	Price	Publications		
101	Master Spring Boot	Srinivas Dande	10000.0	JLC	View Book	Edit Book
102	Master MicroServices	Srinivas Dande	10000.0	JLC	View Book	Edit Book
103	Master Angular	Srinivas Dande	10000.0	JLC	View Book	Edit Book
104	Master React	Srinivas Dande	10000.0	JLC	View Book	Edit Book
105	Master Java Full Stack	Srinivas Dande	10000.0	JLC	View Book	Edit Book



1) index.html

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<title> JLCBookStore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>
<h1 class="text-center"> Welcome to JLCINDIA !!! </h1>
<br/>
<h2 class="text-center"> <a th:href="@{/showBooks}">Show Books</a></h2>

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

</body>
</html>
```

2) BooksList.html

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<title>JLC Bookstore</title>
<link href="webjars/bootstrap/4.3.1/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<div class="container">
<h1 class="text-center"> Welcome to JLCINDIA !!! </h1>

<div class="alert alert-primary" role="alert">
<h2 class="text-center"> Latest Books from Srinivas Dande</h2>
</div>

<br />
<table class="table table-striped table-bordered table-hover"
style="font-size: 20px;">
<thead class="thead-light">
<tr>
<th>Book Id</th>
<th>Book Name</th>
<th>Author</th>
<th>Price</th>
<th>Publications</th>
<th></th>
<th></th>
</thead>
```



```
</tr>
</thead>
<tbody>
<tr th:each="mybook : ${session.MyBooks}">
<td th:text="${mybook.bid}">      </td>
<td th:text="${mybook.bname}"></td>
<td th:text="${mybook.author}"></td>
<td th:text="${mybook.price}"></td>
<td th:text="${mybook.pub}"></td>
<td>
<button type="button" class="btn btn-success btn-lg">View
Book</button>
</td>
<td>
<button type="button" class="btn btn-danger btn-lg">Edit
Book</button>
</td>
</tr>
</tbody>
</table>
</div>

<script src="webjars/jquery/3.4.1/jquery.min.js"></script>
<script src="webjars/bootstrap/4.3.1/js/bootstrap.min.js"></script>

</body>
</html>
```

6) JLCWebConfig.java

```
package com.jlcindia.springboot;

import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class JLCWebConfig implements WebMvcConfigurer{

    @Override
    public void addResourceHandlers(ResourceHandlerRegistry registry) {
        registry.addResourceHandler("/webjars/**")
        .addResourceLocations("classpath:/META-INF/resources/webjars/");
    }
}
```



8)application.properties

server.port=54321

spring.thymeleaf.cache=false

9)pom.xml

<project ...>

<modelVersion>4.0.0</modelVersion>

<groupId>**com.jlcindia.springboot**</groupId>

<artifactId>**demo2-mvc-boot**</artifactId>

<version>1.0</version>

<name>**demo2-mvc-boot**</name>

<description>This is My Spring MVC App with Boot</description>

<parent>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-parent</artifactId>

<version>**2.5.3**</version>

</parent>

<dependencies>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>**spring-boot-starter-web**</artifactId>

</dependency>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>**spring-boot-starter-thymeleaf**</artifactId>

</dependency>

<dependency>

<groupId>org.webjars</groupId>

<artifactId>**bootstrap**</artifactId>

<version>4.3.1</version>

</dependency>



```
<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>myjlcapp</finalName>
<plugins>

<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```